

Calculus: Limits & Derivatives

1. What a Limit Actually Means

A limit describes the value a function approaches as the input approaches some point — it says nothing about the value at that point itself.

- $\lim_{x \rightarrow a} f(x) = L$ means $f(x)$ gets arbitrarily close to L as x gets close to a .
- Limits can exist even where the function is undefined at that exact point.
- One-sided limits (left/right) must agree for the two-sided limit to exist.

2. The Derivative as a Rate of Change

The derivative $f'(x)$ is the limit of the average rate of change as the interval shrinks to zero:

$$f'(x) = \lim_{h \rightarrow 0} [f(x+h) - f(x)] / h$$

Geometrically, this is the slope of the tangent line at x .

3. Core Differentiation Rules

- Power rule: $d/dx[x^n] = n \cdot x^{n-1}$
- Product rule: $(fg)' = f'g + fg'$
- Quotient rule: $(f/g)' = (f'g - fg') / g^2$
- Chain rule: $d/dx[f(g(x))] = f'(g(x)) \cdot g'(x)$

4. Common Mistakes

- Forgetting the chain rule on composite functions.
- Confusing average rate of change with instantaneous rate of change.
- Assuming continuity implies differentiability — it doesn't (e.g. $|x|$ at $x=0$).